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Change Type Analysis for Urban Construction Using Time-Series SAR Data

Boram Lee, *Member, IEEE*, Yoon-Kyung Lee, *Member, IEEE*, and Sang-Wan Kim, *Member, IEEE*

***Abstract*— Monitoring urban changes, particularly large-scale building construction, using satellite imagery is essential for effective urban planning and environmental management. While traditional bitemporal analyses capture the extent of change between two dates, time-series SAR-based approaches enable the detection of temporal patterns leading to the generation of change type classification maps. However, existing time-series change classification methods, which typically categorize changes into broad classes such as step, impulse, cycle, and complex changes, lack the granularity necessary to interpret the underlying physical processes and construction phases that drive urban transformation. This study addresses this limitation by introducing a fine-grained change classification framework specifically designed to enable semantic interpretation of building construction processes. Rather than focused on heuristic class expansion, our framework establishes a physically grounded classification by subdividing conventional categories based on three critical signal characteristics: change direction, duration and periodicity. Using Sentinel-1 SAR GRD time-series data from 2019 to 2023, changes were clustered through a hybrid framework combining likelihood ratio testing and spectral clustering. The classification achieved an overall F1-score of 0.94 on the reference dataset, with particularly strong performance for construction-related change types. The resulting fine-grained classes were subsequently examined in relation to construction-related phenomena using external contextual evidence. Case studies of large-scale apartment development sites further showed that the classified patterns can support construction-related interpretation and spatial analysis of relative construction progression. These semantic associations are demonstrated within the evaluated construction environments.**

***Index Terms*— change detection, construction monitoring, semantic interpretation, spectral clustering, synthetic aperture radar (SAR) time-series**

I. INTRODUCTION

THE detection and monitoring of changes using satellite imagery are crucial for a wide range of applications including urban planning, environmental management, and disaster [1]. Spatial change of urban is typically identified through two-date image analysis, while multitemporal approaches capture long-term changes [2]. Multitemporal change detection not only identifies the presence of change but also reveals patterns of change that are undetectable with two-date change detection. Patterns of change refer to time-series information of intensity at specific locations, which improves our understanding of land cover and land use change [3]. Also, the extraction of temporal patterns allows characterizing what happens within the urban area [4], and it can support more informed urban planning and management [5].

Recent optical remote sensing studies have explored feature-relationship and ensemble-learning approaches for land-use/land-cover classification using multispectral, hyperspectral, and time-series imagery [6]–[8]. These studies show that spatial and temporal relationships can complement conventional pixel-level features in remote sensing classification. Unlike optical satellites, Synthetic Aperture Radar (SAR) enables consistent all-weather observation making it highly effective for reliable time-series change detection [5]. Recent research in SAR time-series change analysis has focused on four main approaches: clustering-based [2], [9] statistical [10], [11], [12] deep learning-based [3], [13] and hybrid [2], [14], [15] classification. The hybrid method combines a statistical approach with a clustering-based classification framework. In this method, the likelihood ratio test is first applied to quantitatively evaluate the probability of change, after which spectral clustering is employed to automatically classify the change types. The change types proposed in previous studies include the unchanged class, impulse change class, cycle change class, step change class, and complex change class [2], [3], [16], [17]. The impulse change class is indicative of fluctuations in backscatter coefficient that occur due to transient events including temporary infrastructure changes and moving objects. The cycle change class represents periodic variations in backscatter coefficient. These variations are typically result of seasonal environmental changes, recurring human activities, or infrastructure usage patterns. The step change class denotes an abrupt and sustained shift in backscatter coefficient, typically caused by long-term structural modifications such as land cover changes, construction, or demolition activities. Complex change refers to a class that is not as clearly defined as the other classes, exhibiting three or more intensity variations.

However, this and subsequent approaches suffer from three critical limitations that restrict their practical utility in real-world urban monitoring applications. First, the coarse categorization scheme which groups all gradual changes into a

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The authors are with Department of Energy Resources and Geosystems Engineering, Sejong University, 05006, South Korea. (brlee254@sejong.ac.kr; ykeunice.lee@gmail.com; swkim@sejong.edu)

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single step class and all temporary changes into a single impulse class fails to distinguish between physically distinct phenomena that occur during urban development. For instance, a step change could represent either permanent building construction (upward backscatter increase) or land clearing (downward backscatter decrease) yet these fundamentally different processes are not differentiated in conventional classifications. Second, existing methods lack the temporal resolution needed to distinguish between short-duration transient events and extended construction phases that span weeks or months. Third, previous studies lacked physical interpretation of classified changes in terms of actual urban development processes. These limitations are particularly problematic in rapidly urbanizing regions such as the Korean Peninsula [18], [19], where South Korea experienced transformation from 20% urbanization in 1965 to over 90% urbanization currently, accompanied by large-scale apartment construction projects and continuous urban expansion [18]. In such contexts, decision-makers require fine-grained spatiotemporal information [20], [21] that can distinguish between different stages of construction, identify areas of active versus suspended construction, and map the spatial progression of development within large project sites.

In this study, a fine-grained change classification framework is presented using time-series Sentinel-1 SAR GRD data. This study focuses on capturing the spatiotemporal patterns of large-scale building construction sites. Building upon the established detection and clustering framework introduced in [2], this study advances time-series change analysis by introducing a fine-grained classification scheme structured around change direction, duration, and periodicity. The principal contribution therefore lies in the construction-oriented semantic interpretation of these fine-grained classes and their application to distinguishing different processes and stages within large scale building-construction sites. For this purpose, the proposed method subdivides conventional change classes into more descriptive subtypes to enable representation of construction-driven changes, temporary human activity patterns, and stabilization behaviors. Accordingly, the study makes three specific contributions: (1) subdivision of change classes based on explicit temporal characteristics rather than arbitrary expansion of categories, (2) performance evaluation using an expert-annotated dataset independently interpreted from the original SAR time-series; and (3) site-specific semantic analysis linking the resulting patterns to construction stages using optical imagery, building footprints, and official construction records. To evaluate the temporal-pattern classification performance of the proposed framework, we constructed an expert-annotated reference dataset from Sentinel-1 SAR time-series acquired over the study area in South Korea. This dataset was used exclusively to calculate performance metrics by comparing the algorithm-derived classes with the expert-interpreted temporal-pattern labels.

II. STUDY AREA AND DATA

A. Study Area

This study analyzed change patterns across a variety of contexts, focusing on urban areas (Fig. 1). The study area covers urban and peri-urban environments in South Korea. Although diverse temporal changes occur across the scene, the construction-related analysis focuses on three large scale apartment development sites that underwent substantial construction during the 2019–2023 observation period (Fig. 1). The false color composite was used to identify locations exhibiting pronounced temporal changes, and the boxed regions indicate the construction sites examined in detail.

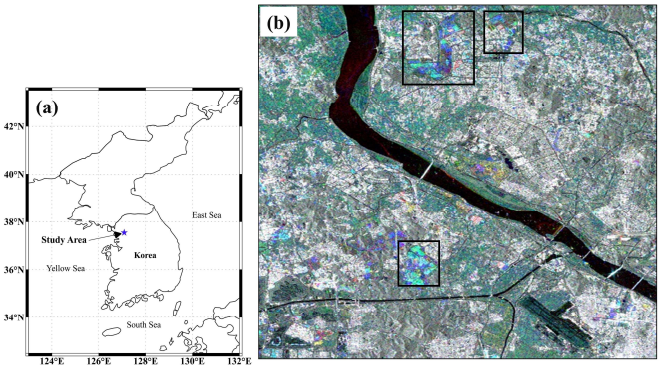


Fig. 1. (a) Location of the study area. (b) False color composite of the study area (R: 20190109, G: 20210522, B: 20231226). The boxed regions indicate three sites where large-scale apartment construction took place during the study period, representing areas with significant changes.

B. Sentinel-1 SAR Data

The Sentinel-1 SAR data used in this study (Table 1) was obtained from the cloud-based platform, Google Earth Engine (GEE) [22]. GEE is a platform that operates in the computational cloud, which is designated mainly for analyzing large sets of remote sensing and vector data. The dataset is Sentinel-1 Level-1 Ground Range Detected (GRD) data. Preprocessing of the data was performed using the Sentinel-1 Toolbox, which included thermal noise removal, radiometric calibration, and orthorectification. The dataset is updated daily, with new assets typically becoming available within two days of acquisition. The spatial resolution of the Sentinel-1 data is available at 10 meters, 25 meters, and 40 meters. For this study, data with a spatial resolution of 10 meters was downloaded. The detailed description of the data collected for the two study areas is presented in Table 1. The data for both study areas consists of images captured on ascending orbits at 09:00 UTC.

TABLE I
DESCRIPTION OF SENTINEL-1 SAR DATASET

Acquisition date	2019/01~2023/12
Number of scenes	133
Acquisition time	09h (UTC)

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Sensor mode	IW mode
Orbit pass	Ascending
Polarimetry	VV
Product	Ground Range
Pixel spacing	10m x 10m

C. Labeled Data for Algorithm Experimentation

The accuracy of the proposed classification framework was evaluated using a manually annotated reference dataset representing eight temporal change types. Previous studies have commonly used synthetic time-series data to evaluate change-pattern classification methods [2], [3], [17]. Although synthetic data provide clearly defined change patterns, they may not fully represent the complex temporal variability, residual noise, and ambiguous class boundaries observed in real SAR time-series. Therefore, this study constructed the reference dataset directly from Sentinel-1 SAR backscatter time-series acquired over the study area.

Candidate locations were distributed as evenly as possible across the study area to capture the spatial variability of the observed temporal patterns. Two researchers independently assigned preliminary labels by visually examining each time-series trajectory. The annotation criteria included shifts in the mean backscatter level, the occurrence and direction of abrupt changes, the number and duration of temporal peaks, and the presence of repeated fluctuations. The independently assigned labels were subsequently cross-reviewed, and disagreements were resolved through discussion to determine the final reference label. For periodic patterns that could not be clearly distinguished through visual inspection alone, Fourier analysis was used only as an auxiliary check to examine the presence and number of dominant periodic components.

The final reference dataset comprised 1,462 time-series samples: unchanged (n = 227), complex change (n = 134), down step (n = 163), up step (n = 185), spike impulse (n = 209), long impulse (n = 178), regular cycle (n = 174), and irregular cycle (n = 192) (Fig. 2). All samples were extracted from distinct spatial coordinates. A within-class temporal profile redundancy analysis was additionally conducted to determine whether the dataset was dominated by repeated or nearly identical temporal trajectories. These manually assigned labels were generated without reference to the outputs of the proposed algorithm and were used to evaluate agreement between the algorithm-derived classes and expert-interpreted SAR temporal patterns.

To examine whether the reference dataset contained repeated or nearly identical temporal profiles, Pearson correlation coefficients were calculated for all possible pairs of time-series within each class. Pairs with r >= 0.95 were considered highly similar, and their proportion among all possible pairs was calculated for each class. Across the eight classes, highly similar pairs accounted for 0.40–10.85% of all within-class pairs. This result indicates that the reference dataset was not dominated by repeated or nearly identical temporal profiles and contained heterogeneous temporal behaviors within the target scene.

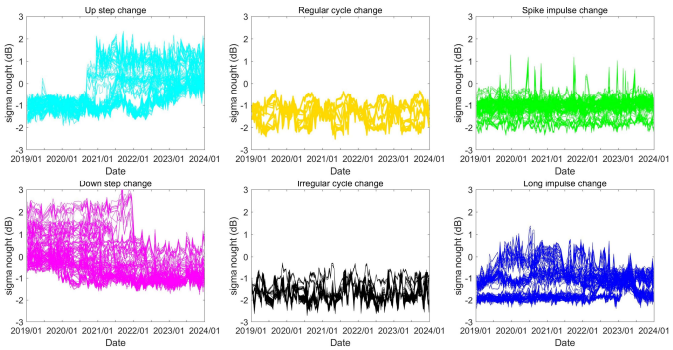


Fig. 2. Time-series plots of labeled change types derived from Sentinel-1 SAR data. Each plots represent classified change types: down step, up step, spike impulse, long impulse, regular cycle, and irregular cycle.

III. METHODS

The methodology consists of spatiotemporal speckle filtering, change detection, and change type classification to derive a classification map (Fig. 3). Change detection methods are generally based on parametric and non-parametric approaches to test for changes in time-series. In this study, spectral clustering method within a parametric framework is used for time-series change detection [2]. This algorithm, first introduced by [23], applies a similarity matrix constructed with the parametric likelihood ratio test instead of calculating the conventional distance-based similarity matrix. By leveraging spectral clustering's eigenspace-based dimensionality reduction, enhances the separation of latent structures and facilitates the robust identification of diverse change types within complex time-series data.

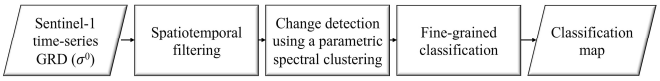


Fig. 3. Flowchart of the proposed methodology.

A. Simple spatiotemporal filtering method

One of the most important problems of SAR data is speckle noise involved in coherent acquisition systems, which reduces the accuracy of SAR change detection [24]. Therefore, it is necessary to properly remove speckle noise before applying a change detection algorithm. In existing SAR change detection studies, various spatial filtering techniques have been proposed to reduce speckle noise, but they are based on local statistics of a single image, which limits the preservation of time-series patterns. The rise of multi-temporal SAR-based change detection has highlighted the need for spatiotemporal filtering that considers both temporal and spatial characteristics, and techniques such as NL-SAR and 2S-PPB have the limitation that filtering is performed even in the region containing the change, which suppresses the real change signal [25]. To overcome this problem, we applied the

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DespecKS algorithm [26], which is based on the two-sample Kolmogorov-Smirnov test, to extract pixels with statistical similarity in space and time, effectively removing speckle noise in non-changing regions while preserving the change signal. In this study, pixels that meet the 95% significance level were assigned a binary weight of one and subsequently averaged. This setting was initially selected as a practical compromise between speckle suppression in stable areas and the preservation of subtle temporal signatures in active construction zones. Preliminary analyses suggested that the more restrictive 90% confidence setting selected fewer homogeneous neighbors, resulting in weaker speckle suppression. We subsequently evaluated this empirical choice at all 1,462 validation locations using confidence levels of 90%, 95%, and 99%. Overall accuracy and macro F1-score varied by no more than 0.55 and 0.54 % respectively, and fewer than 1.6% of the samples changed their predicted class. Thus, the 95% setting provides a reasonable intermediate operating point, while the limited performance variation indicates that the reported results are robust to this parameter.

B. Spectral clustering approach for time-series data

Among the various clustering methods [15], [27-30], spectral clustering is known to be particularly suitable for complex graph-based data, such as time-series data [31]. While spectral clustering is a well-established unsupervised classification method in the broader domain of image analysis, its application within a parametric framework for SAR time-series change detection was prominently advanced by [2]. In this study, the similarity matrix for change detection is constructed using the Likelihood Ratio Test (LRT) between temporal points, referred to as the Change Criterion Matrix (CCM). Assuming that SAR backscatter intensity follows a Gamma distribution [32], the CCM is formulated as follows:

$$\mathrm{CCM} = \begin{bmatrix} \mathrm{LRT}(y_1, y_1) & \cdots & \mathrm{LRT}(y_1, y_n) \\ \vdots & \ddots & \vdots \\ \mathrm{LRT}(y_n, y_1) & \cdots & \mathrm{LRT}(y_n, y_n) \end{bmatrix}$$

$$, \text{where } \mathrm{LRT}(y_i, y_j) = \frac{p(y_i, y_j | u_{ij})}{p(y_i | u_i) p(y_j | u_j)} \qquad (1)$$

Here, y1 and y2 represent the backscattering coefficient value at two temporal points that include noise and u12 represents the noise-free reflectance assuming no change between the two temporal points. Following the construction of the similarity matrix, the algorithm implements the Laplacian transformation on the similarity matrix, followed by an eigenspace decomposition. The resulting eigenvalues are then arranged in ascending order and the differences between adjacent eigenvalues are calculated. The index with the maximum difference is then used to determine the optimal number of clusters. Since the eigengap in the spectral clustering algorithm is sensitive to relative pattern variations in time-series data, even residual speckle noise can lead to an overestimation of the cluster number ($p \geq 2$). To address this, we incorporated a post-processing step using a conservative Coefficient of Variation (CV) threshold of 0.65. A local sensitivity analysis was conducted by varying the CV threshold from 0.55 to 0.75 in increments of 0.05 while fixing all other parameters. The resulting overall accuracy, macro F1-score varied by no more than 0.27 and 0.25 %, respectively. Furthermore, fewer than 0.83% of the validation samples changed their predicted class relative to the default threshold of 0.65, demonstrating limited sensitivity to this empirical parameter.

This integrated clustering approach provides the foundational cluster labels for the subsequent fine-grained change classification. The comprehensive workflow of the algorithm including this post-processing step is detailed in Fig. 4.

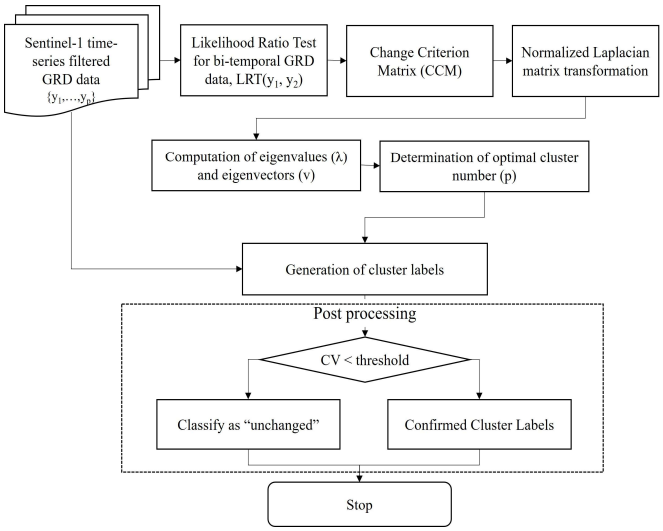


Fig. 4. Flowchart of the spectral clustering algorithm.

C. Fine-grained classification of change classes

Unlike natural changes, the complexity of change patterns in urban areas arises primarily from human-induced activities. This complexity imposes limitations on conventional change classification methods, making it difficult to precisely identify the underlying causes of changes. The expected types of changes under various urban scenarios can be summarized as illustrated in Fig. 5. Human-induced activities can generate diverse temporal responses in SAR time series. The taxonomy in Fig. 5 summarizes potential interpretations of these temporal pattern classes. The semantic analysis in this study is restricted to large scale building construction sites. Accordingly, permanent structure formation, temporary site activity, and post-construction clearance are examined as context-specific interpretations.

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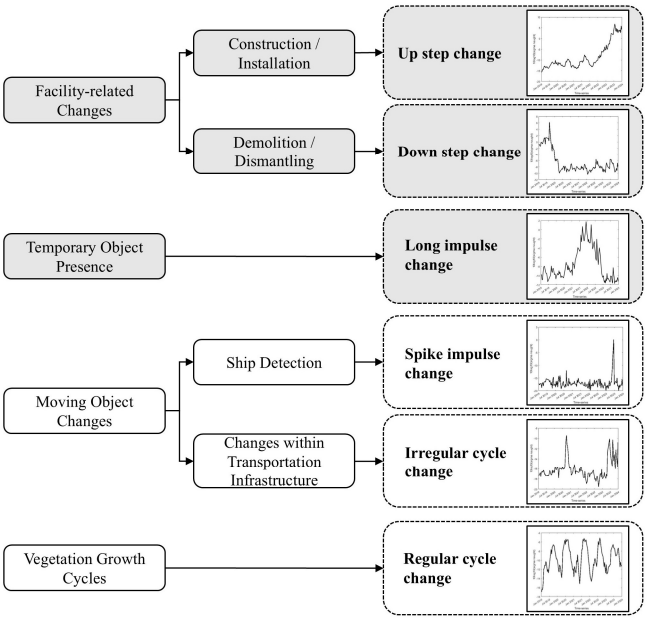


Fig. 5. Conceptual relationships between SAR time-series change classes and potential urban-change processes. Construction-related interpretations evaluated in this study are highlighted.

To address this limitation, the conventional change classes were further subdivided using change direction, duration and periodicity. Once cluster labelling is complete, change types are determined based on the temporal arrangement of the clusters. In this section, we propose extending the existing classification of change types by incorporating three key considerations: change direction, periodicity, and duration. Fig. 6 illustrates the algorithm flowchart. Classification is performed based on two parameters: the number of clusters (n) and the number of changes (k). If the number of clusters (n) is 1, the data is classified as unchanged. When n=2, classification is further refined based on the number of changes (k), leading to step, impulse or cycle change classes. The classification process is detailed in Fig. 6. If k=1 (one change event), the classification is determined by comparing the mean values of the two clusters before and after the change. If the latter cluster has a higher mean value, it is categorized as up step while a lower mean value results in a down step class. For k=2 (two change events), the signal initially changes but later returns to its original cluster. In this case, the duration of change (Δt) is calculated. If Δt=1, the change is classified as spike impulse, otherwise it is categorized as long impulse change. Lastly, if $k \geq 3$ (three or more change events), the data represents repeated changes within the same cluster, indicating a cyclic pattern. To distinguish the periodicity, the periodogram is applied to transform the time-series backscatter into the frequency domain. If a dominant peak is identified, the change is classified as regular cycle change; if multiple peaks are present, it is classified as irregular cycle change. In the case of n=3, we defined an others class to represent complex changes; however, this class was excluded from the subsequent analysis.

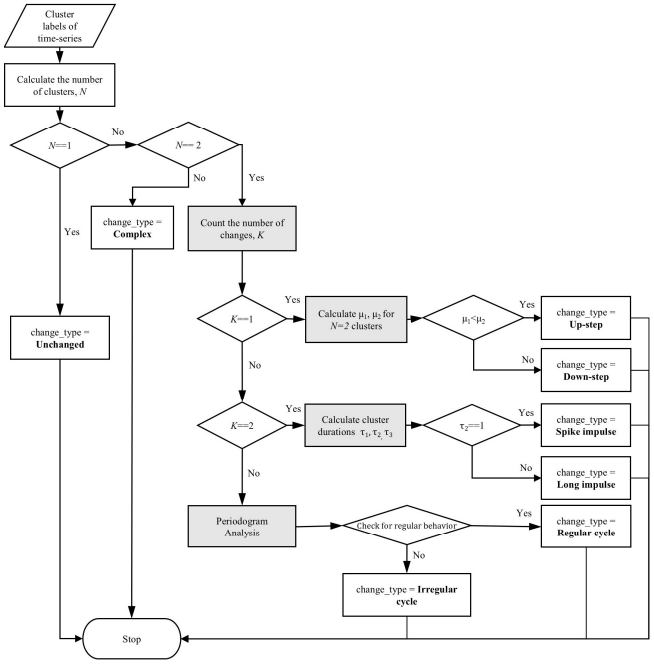


Fig. 6. Flowchart for SAR-based classification of change types.

IV. RESULTS AND DISCUSSIONS

A. Analysis of change classification accuracy

The classification performance was evaluated at two levels. First, the distinguishability of the proposed eight fine-grained change classes was assessed using the original reference labels. Second, to enable comparison with an existing coarse-grained change-analysis method, the eight classes were mapped to the four temporal pattern superclasses shared with NORCAMA [2]: unchanged, step, impulse, and cycle change. Specifically, the up step and down step classes were merged into step change; spike impulse and long impulse were merged into impulse change; and regular cycle and irregular cycle were merged into cycle change. For this comparison, a baseline was constructed using the 2S-PPB multitemporal despeckling method adopted in NORCAMA [2]. The 2S-PPB-based baseline and the proposed method were evaluated using the same input SAR observations, reference samples, and superclass definitions.

Following this validation strategy, a confusion matrix was analyzed to evaluate classification performance on the scene-specific reference dataset (Table 2). The diagonal elements indicate correctly classified samples, while the off-diagonal elements represent misclassified cases. The matrix shows precision and recall values exceeding 0.95 for most classes. In particular, the up step and long impulse change classes achieved F1-scores of 1.00 and 0.97, respectively. The others class showed relatively lower recall compared to its precision due to misclassification of complex change samples. However, as this class is not the primary focus of the study, this limitation is of minor concern. This limitation originates from

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the spectral clustering algorithm itself, whereby samples are assigned to the others category when the number of clusters is incorrectly estimated as three. The classification maintained precision and recall above 0.95 for construction-related change types, including step and impulse changes.

TABLE II

CONFUSION MATRIX WITH PERCENTAGE VALUE OF FINE-GRAINED CLASSIFICATION RESULT USING LABELED DATA

True \ Predict		Step		Impulse		Cycle		Others	Unchanged	Recall (%)
		Down	Up	Spike	Long	Regular	Irregular			
Step	Down	162	0	0	1	0	0	0	0	**0.99**
	Up	0	180	0	0	3	0	2	0	**0.97**
Impulse	Spike	0	0	200	0	5	0	1	3	**0.96**
	Long	0	0	1	169	0	0	7	1	**0.95**
Cycle	Regular	0	0	0	0	162	2	0	12	**0.93**
	Irregular	0	0	0	0	0	189	3	0	**0.98**
Others		1	7	0	5	12	4	100	5	**0.75**
Unchanged		0	0	0	0	5	1	0	221	**0.97**
Precision (%)		**0.99**	**0.96**	**1.00**	**0.97**	**0.87**	**0.97**	**0.88**	**0.91**	

Table III presents the comparison between the 2S-PPB baseline and the proposed method. For the four shared superclasses, the 2S-PPB-based baseline achieved an overall accuracy of 0.8931 and a macro F1-score of 0.9014, whereas the proposed method achieved values of 0.9684 and 0.9711, respectively. The class-wise F1-scores for step, impulse, and cycle were 0.9208, 0.8528, and 0.8962 for the 2S-PPB-based baseline and 0.9814, 0.9738, and 0.9410 for the proposed method, respectively. Both methods achieved overall accuracy above 0.89, while the proposed method yielded higher class-wise F1-scores under the evaluated conditions. Examination of the confusion matrix showed that some others samples containing relatively weak or complex fluctuations were classified as step or impulse after 2S-PPB processing, thereby increasing the number of false positives and reducing the precision and F1-scores of these classes. Although 2S-PPB effectively suppressed speckle and maintained generally high classification accuracy, its temporal smoothing occasionally attenuated short-duration or spatially localized variations, producing simplified trajectories that resembled step or impulse patterns.

TABLE III

SUPERCLASS-LEVEL CLASSIFICATION PERFORMANCE OF THE 2S-PPB-BASED-BASELINE AND THE PROPOSED METHOD

Method		2S-PPB-based baseline	Proposed method
Overall accuracy		0.893	0.968
Macro F1-score		0.901	0.971
F1-score	Step	0.921	0.981
	Impulse	0.853	0.974
	Cycle	0.896	0.941

Class dependent performance has also been reported in previous evaluations of NORCAMA. Using realistic TerraSAR-X time-series data, [5] reported NORCAMA F1-scores of 0.7241, 0.6121, and 0.3858 for the step, impulse, and cycle classes, respectively. These values are not directly comparable with the present results because of differences in the sensor, study area, temporal sampling, reference data, and detailed processing conditions. Nevertheless, the previously reported results provide external context showing that the classification performance of NORCAMA can vary substantially according to temporal-pattern class. The present experiment similarly demonstrates the importance of preserving class discriminative temporal variations during multitemporal despeckling.

B. Change classification map

The study area is characterized by a variety of land cover and land use types such as cropland, a river estuary, an airport, and large-scale apartment construction sites. This setting allows for the identification of distinct spatial patterns of change classes in the change classification map (Fig. 7). The analysis presented here focuses on the spatial patterns observed at large-scale apartment construction sites. The five year SAR time-series captured these sites as spatially coherent clusters containing up step, down step, and long impulse changes.

As shown in the black boxes within the overview map (Fig. 7), which represent construction sites (a–c), areas characterized by the combined presence of up step, long impulse, and down step changes were identified as large-scale apartment construction sites. These boxed regions, enlarged in the detailed subfigures (Figs. 7a–c), shows that up step changes were primarily detected at building sites, while long impulse changes appeared in the spatial intervals between areas classified as up step changes. Within the enlarged construction sites shown in Figs. 7a–7c, up step changes were concentrated around building locations, while long impulse and down step changes were also observed in adjacent site areas. These results demonstrate that a single construction site may contain multiple temporal-pattern classes rather than a spatially uniform step change response.

C. Case-Study-Based Semantic Interpretation of Urban Construction Changes

As shown in the change classification map in Fig. 7c, the apartment-development areas exhibited not only up step changes but also distinct down step and spatially coherent long-impulse changes. This coexistence of different change types within the same spatial domain implies that multiple change mechanisms operated concurrently during the area's development process, reflecting the complexity of urbanization. To relate these SAR-derived patterns to actual construction progress, two representative apartment-development sites within the area shown in Fig. 7c were further examined.

Fig. 8a provides an overview of the broader Unjeong New Town development area, while Figs. 8b and 8c present enlarged views of the two selected sites. According to public building records, the construction periods of the sites shown in Figs. 8b and 8c extended from June 3, 2021, to February 29, 2024, and from December 13, 2017, to July 22, 2020, respectively. Thus, the site in Fig. 8c had entered the post-

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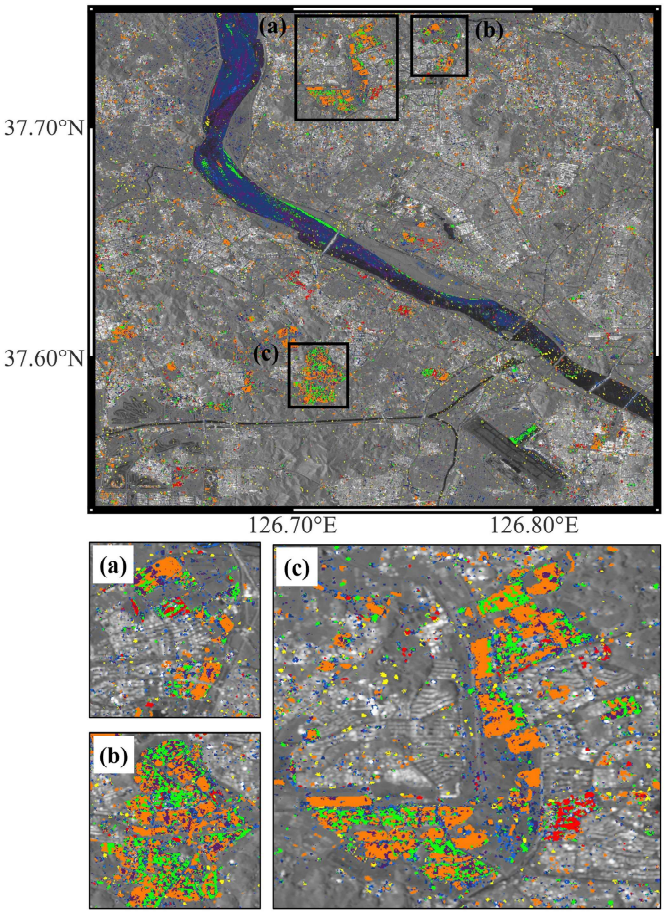


Fig. 7. Change classification map of the study area. The colors illustrated in (a) represent different change types: black for no change, red for down step changes, orange for up step changes, yellow for spike impulse changes, green for long impulse changes, blue for regular cycle changes, navy for irregular cycle changes and purple for complex changes. Subfigures (a–c) show where large-scale apartment construction took place during the study period.

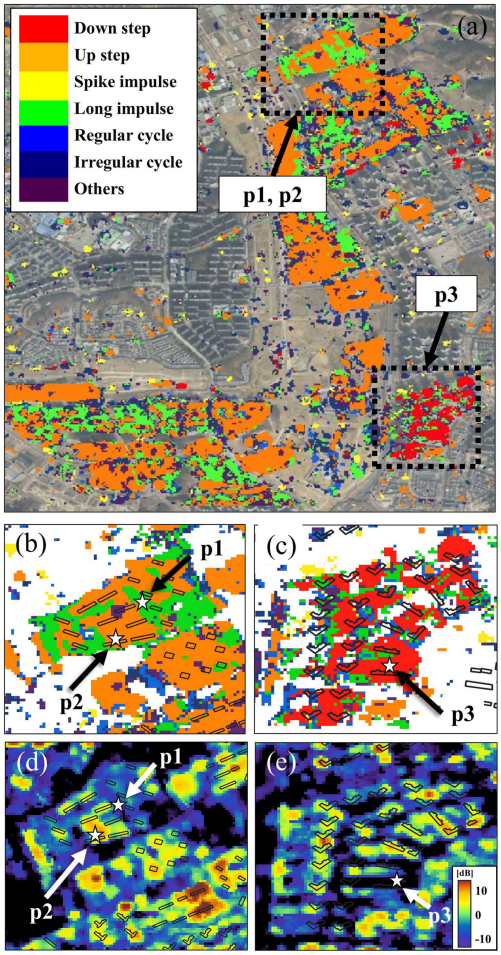


Fig. 8. (a) A change classification map showing the locations of p1, p2, and p3, with dashed boxes indicating the zoomed-in regions displayed in (b–e). (b, c) Enlarged classification results for the areas containing p1–p2 and p3, respectively. (d, e) Corresponding SAR backscattering coefficient images (in dB) for the same regions at the final observation date (26 December 2023), with the locations of p1, p2 and p3 marked.

construction stage by July 2020, whereas the site in Fig. 8b remained in the pre-approval stage at the time of the final SAR acquisition on December 26, 2023. These independently documented construction timelines demonstrate that the two sites represented different development stages during the observation period.

Building footprints obtained from OpenStreetMap (OSM) were overlaid on the classification results as an independent spatial reference. Across the two development sites, most areas overlapping the building footprints exhibited clear up step patterns, whereas down step and long impulse changes were broadly associated with outside the footprints, particularly in inter-building spaces, although exact pixel-level correspondence was not observed because of SAR geometric distortions.

Taken together with the documented approval dates and footprint, these spatial relationships indicate that the mapped classes capture different temporal responses occurring within apartment-development sites. The contrast between Fig. 8c, which was completed during the earlier part of the observation period, and Fig. 8b, which remained in the pre-completion stage until after the final SAR acquisition, further shows that differences in construction stage can produce different temporal-change patterns. In addition, the coexistence of multiple change types within each site reflects spatially heterogeneous activities occurring during construction rather than a single uniform construction signal.

Further investigation combining high-resolution Google Earth imagery and time-series backscatter coefficient at three points was conducted to gain a better understanding of the change mechanisms (Fig. 9). Google Earth imagery confirmed that apartment complexes were constructed during the study period at the p1 and p2 sites. p2 corresponds to the building, while p1 is located in the space between buildings, where construction materials were observed in the February 2024 image. The time-series backscatter coefficient at p2 remained low during the early observation period, increased during the

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construction phase, and then slightly decreased and stabilized after completion, indicating a stable scatterer formed by the completed structures. In contrast, p1 exhibited a temporary increase in backscatter coefficient during the same period as adjacent building construction, followed by a return to pre-construction levels after project completion. This suggests that the p1 did not contain permanent structures and that backscatter signal was caused by temporary materials, steel frameworks, or equipment. Similarly, p3 is located between buildings, where high initial backscatter coefficient gradually decreased, reflecting the removal of temporary structures as construction progressed. These results demonstrate that differences in construction stages within the observation period can lead to variations in classified change types. The proposed fine-grained classification approach therefore provides a more detailed interpretation of urban construction processes than conventional methods that simply associate step changes with building construction.

However, it is important to note that the spatial distribution of change patterns may not perfectly align with the exact physical footprints of individual buildings. This discrepancy arises because SAR geometric distortions, such as layover and foreshortening, are governed not only by the observation geometry but also by the complex interaction between the three-dimensional incident beam angle (comprising azimuth and elevation components) and individual building orientations [~~31~~33]. Furthermore, as buildings grow taller during the construction phase, the area affected by layover can gradually expand over time [34], [35], which presents a fundamental challenge for SAR-based change classification. Under idealized ground-range geometry, the layover extent of a vertical structure can be approximated as ($L = H \cot\theta$), where (H) is the building height and (θ) is the incidence angle [36]. Accordingly, for the nominal Sentinel-1 IW incidence-angle range of 29.1°~46.0°, the idealized ground range layover extent corresponds to approximately 1.80–0.97 times the building height. Thus, construction related height growth can shift the observed backscatter response toward the sensor by a distance comparable to or greater than the building height.

Under the evaluated ascending orbit configuration, the temporal classes could be distinguished at the construction-block scale when building orientation and inter-building spacing provided favorable geometric conditions. The interpretation is expected to be more reliable when construction-related changes extend over multiple pixels, form spatially coherent clusters, and the range-projected spacing between buildings is sufficiently large relative to the expected layover extent. Conversely, the classes may become spatially ambiguous for subpixel, building orientations approaching weak double-bounce configurations, and densely built areas where layover and radar shadow from adjacent structures overlap. However, all experiments were conducted using a single ascending orbit Sentinel-1 viewing geometry, and transferability to descending orbit observations, different incidence angles, or differently oriented urban structures was not evaluated. The results should thus be interpreted as site-level construction monitoring under the evaluated geometry rather than as geometrically invariant building-footprint classification.

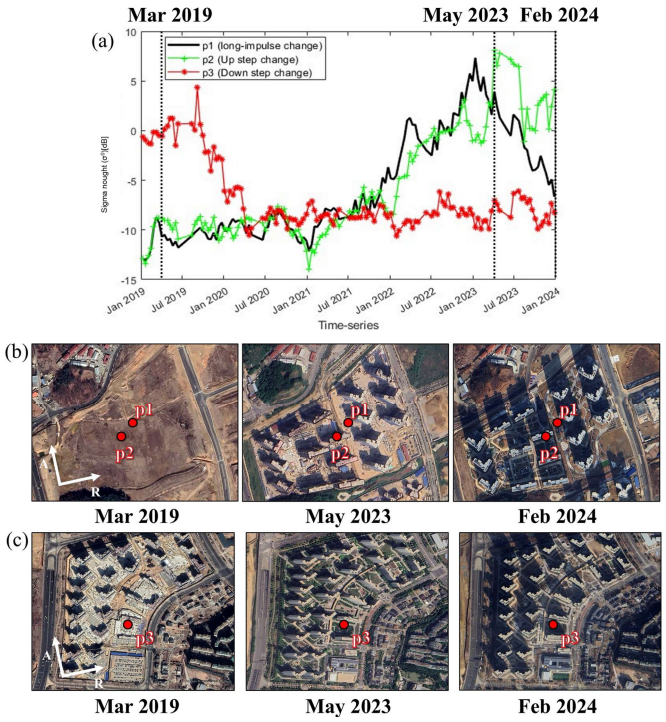


Fig. 9. Semantic interpretation of SAR backscattering coefficient trajectories and optical validation across diverse construction scenarios: (a) Time-series backscatter coefficient profiles for p1 (long impulse change; temporary storage), p2 (up step change; permanent building) and p3 (down step change; temporary storage removal). (b) Optical validation for p1 and p2, showing the progression from pre-construction to the structural phase. (c) Optical validation for p3, capturing the transition from an active construction site with temporary facilities to the final completion state following their removal. Red dots indicate the exact pixel locations and labels A and R denote the Azimuth and Range directions, respectively. The alignment of temporal markers (Mar 2019, May 2023 and Feb 2024) demonstrates the method's capability to correlate latent SAR signatures with specific mechanism.

The long impulse changes observed at construction site as illustrated in Fig. 9 can be interpreted as temporary storage of construction materials. To support this interpretation and evaluate the overall spatiotemporal trends, the spatial distribution of spike and long impulse changes in the corresponding area was visualized (Fig. 10a). The result clearly shows that these changes are classified as long impulse types within the spaces between high-backscatter building structures, indicating the localized and temporary nature of material storage during the construction process. In contrast, spike impulses exhibit a scattered and non-clustered distribution across the study area. This spatial disparity supports the interpretation of long impulse changes are tied to systematic and spatially coherent construction activities, spike impulses represent stochastic and isolated transient events.

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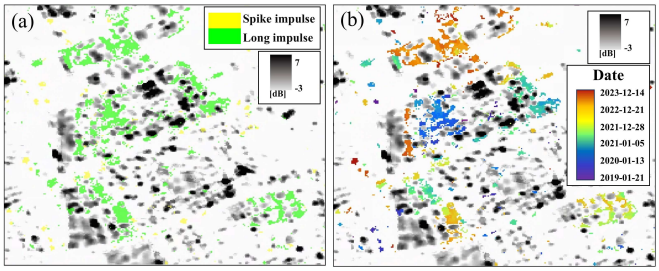


Fig. 10. (a) Classification map of impulse changes, overlaid on a SAR backscatter coefficients (sigma naught, σ^0) expressed in decibels [dB] image. (b) Spatial distribution of the average date of long impulse changes, visualized in color, overlaid on a SAR backscatter coefficients

Based on this evidence, the analysis of the timing and duration of the changes enables an estimation to be made of the construction period and the stage. Fig. 10b visualizes the average value of the date for long impulse change pixels. The blue color represents the past and the red color represents the recent changes. The analysis shows that within each building cluster, pixels with similar mean dates of change are clustered together, indicating that the spatial extent of each cluster is clearly distinct. These differences enable us to estimate the construction sequence for each cluster and provide detailed understanding of the construction process based on the results of the change type classification.

Beyond the superclass level performance demonstrated through comparison with the 2S-PPB-based baseline, the proposed fine grained classification provides additional information for interpreting spatially heterogeneous changes within construction sites. Conventional step change detection would identify the entire construction area as "changed" but cannot distinguish building locations (p2, up step) from ground clearance zones (p3, down step) Furthermore, as illustrated in Fig. 10a, whereas a mixture of spike and long impulse changes might typically be interpreted as generic transient human activity, the proposed fine-grained classification reveals that the long impulse changes closely associated with building construction exhibit distinct spatial clustering. This spatial consistency allows for a clear differentiation of construction-driven signals from the isolated occurrences of spike impulses.

Although the reference dataset showed a low prevalence of highly correlated temporal profiles, all 1,462 samples were extracted from the same Sentinel-1 image stack and therefore shared the same sensor, acquisition geometry, observation dates, and regional conditions. Validation using independent SAR stacks acquired over different urban environments remains necessary to establish broader generalizability.

V. Conclusion

Conventional SAR time-series change detection methods have a fundamental limitation. While they can successfully identify when and where changes occurred, they cannot interpret what physical processes caused those changes [37]. A detected step change could represent permanent building construction, land clearing or vegetation removal. These are physically distinct phenomena that require different management responses, yet existing coarse-grained classifications conflate them into a single, undifferentiated category. This ambiguity can limit their application to construction-oriented interpretation.

This study introduced a fine-grained SAR temporal pattern classification framework based on change direction, duration, and periodicity. Applied to Sentinel-1 time-series data from South Korea, the framework achieved an overall F1-score of 0.94 for the defined temporal-pattern classes. Construction-site case studies further demonstrated how up step, long impulse, and down step patterns can support the interpretation of permanent structure formation, temporary site activity, and clearance processes within the evaluated large-scale apartment developments.

The semantic interpretation capability transforms abstract change patterns into detailed description of construction-related temporal behavior at the site scale. Analysis of building construction sites demonstrated that up step changes correspond to permanent structure formation, long impulse changes indicate temporary construction activity zones and material storage, and down step changes mark completion phases or infrastructure clearance. By mapping the temporal progression of long impulse changes across construction sites, project timelines can be spatially delineated and construction phases identified within individual building clusters. The inclusion of details such as the specific construction work that took place, when and where it occurred, allows for direct application in construction progress monitoring, ensuring compliance and estimating development timelines.

The main contribution of this work is the transition from coarse change categories to a more detailed description of construction related temporal behavior at the site scale. Rather than uniquely identifying a physical activity from SAR data alone, the framework supports semantic interpretation when the classified temporal patterns are combined with independent contextual evidence. Mapping the timing of long impulse changes also provides information on the relative progression of construction among building clusters.

To further advance this framework, future research should incorporate interferometric phase information and high-resolution Digital Elevation Models (DEMs) to conduct a quantitative assessment of how geometric distortions influence change classification results during both construction and demolition phases. The integration of these data could support quantitative assessment of geometric effects and improve interpretation under complex urban conditions. Although consistent temporal pattern classes were observed across the evaluated building clusters, further analysis is required to understand the classification behavior for subpixel changes smaller than the 10 meter spatial resolution of Sentinel-1. Building on these findings, future research should extend this semantic interpretation paradigm to other urban change processes, including demolition and infrastructure development, while integrating data sources to further enhance interpretation accuracy and expand the range of semantically mapped phenomena. Also, future research should evaluate the framework using independent sites, diverse urban morphologies, and different acquisition geometries, together with corresponding ground-reference data.

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Boram Lee (Member, IEEE) received the B.S., M.S., and Ph.D. degrees in geoinformation engineering (remote sensing) from Sejong University, Seoul, South Korea, in 2011, 2013, and 2025, respectively.

From 2011 to 2017, she was a Researcher with the Korea Ocean Satellite Center, Korea Institute of Ocean Science and Technology (KIOST), South Korea. From 2020 to 2023, she

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was an Industrial Investigator (Grade 6) with the Korea Coast Guard, South Korea. Since September 2025, she has been a Researcher, Sejong University. Her research interests include change detection algorithm development using SAR data, time-series analysis, marine environmental monitoring, and optical satellite data preprocessing and calibration.



Yoon-Kyung Lee (Member, IEEE) received the B.S. degree in life sciences from Catholic University of Korea, South Korea, in 2001, and the M.S. and Ph.D. degrees in earth system sciences from Yonsei University, South Korea, in 2004 and 2012, respectively.

From 2012 to 2021, she was a Researcher with the Korea Ocean Satellite Center, Korea Institute of Ocean Science and Technology (KIOST), South Korea. Since 2022, she has been a Researcher, Sejong University, South Korea. Her research interests include SAR data processing, ocean satellite remote sensing, maritime vessel detection and tracking using multi-source data fusion, nighttime satellite imagery processing.



Sang-Wan Kim (Member, IEEE) received the B.S. degree in geology from Yonsei University, South Korea, in 1992, and the M.S. and Ph.D. degrees in earth system sciences from Yonsei University in 1998 and 2004, respectively.

From 2004 to 2007, he was a Research Fellow with the University of Miami, Miami, FL, USA. Since 2007, he has been a Professor with the Department of Energy Resources and Geosystems Engineering, Sejong University, South Korea. His research interests include SAR interferometry, InSAR time-series analysis in urban area and geohazard monitoring, and nighttime remote sensing using VIIRS DNB imagery

Since 2020, he has served as the Chair of the Korea Chapter of the IEEE International Geoscience and Remote Sensing Symposium (IGARSS).

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